# Predicting Formation Energy for Fast Materials Screening

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Using ML to approximate DFT formation energy for perovskites

# *CRISP-DM reference model:* Generic tasks (**bold**) and outputs (*italic*)

| Business Understanding | Data Understanding | Data Preparation | Modeling | Evaluation | Deployment |
|---|---|---|---|---|---|
| **Determine Business Objectives**<br>*Background*<br>*Business Objectives*<br>*Business Success Criteria*<br><br>**Assess Situation**<br>*Inventory of Resources*<br>*Requirements, Assumptions, and Constraints*<br>*Risks and Contingencies*<br>*Terminology*<br>*Costs and Benefits*<br><br>**Determine Data Mining Goals**<br>*Data Mining Goals*<br>*Data Mining Success Criteria*<br><br>**Produce Project Plan**<br>*Project Plan*<br>*Initial Assessment of Tools and Techniques* | **Collect Initial Data**<br>*Initial Data Collection Report*<br><br>**Describe Data**<br>*Data Description Report*<br><br>**Explore Data**<br>*Data Exploration Report*<br><br>**Verify Data Quality**<br>*Data Quality Report* | **Select Data**<br>*Rationale for Inclusion/ Exclusion*<br><br>**Clean Data**<br>*Data Cleaning Report*<br><br>**Construct Data**<br>*Derived Attributes*<br>*Generated Records*<br><br>**Integrate Data**<br>*Merged Data*<br><br>**Format Data**<br>*Reformatted Data*<br><br>*Dataset*<br>*Dataset Description* | **Select Modeling Techniques**<br>*Modeling Technique*<br>*Modeling Assumptions*<br><br>**Generate Test Design**<br>*Test Design*<br><br>**Build Model**<br>*Parameter Settings*<br>*Models*<br>*Model Descriptions*<br><br>**Assess Model**<br>*Model Assessment*<br>*Revised Parameter Settings* | **Evaluate Results**<br>*Assessment of Data Mining Results w.r.t. Business Success Criteria*<br>*Approved Models*<br><br>**Review Process**<br>*Review of Process*<br><br>**Determine Next Steps**<br>*List of Possible Actions*<br>*Decision* | **Plan Deployment**<br>*Deployment Plan*<br><br>**Plan Monitoring and Maintenance**<br>*Monitoring and Maintenance Plan*<br><br>**Produce Final Report**<br>*Final Report*<br>*Final Presentation*<br><br>**Review Project**<br>*Experience Documentation* |

# Business Understanding: Why We Care

## Material Discovery Process

**Property Definition** → **Candidate Materials** → **Stability Check**

Define the properties
1. Durability
2. Heat transfer
3. Is it catalyst
4. Transparency
5. Conductivity
….

Choose the candidates analytically, based on the knowledge, chemistry laws, math.

Check the stability. "**Is it possible to synthesize and use such materials?**".

- Materials discovery has a **combinatorial search space** (tons of candidate perovskites).
- Formation energy is a key proxy for **stability / synthesizability**.
- Ground truth via **DFT (Density Functional Theory)** is accurate but expensive → slows iteration.
- Business/scientific value: enable **rapid screening** and prioritize DFT only for top candidates.

# Business Understanding: Objective & Hypotheses

## Hypothesis

- Adding **structure-derived features** should drastically improve over composition-only models.
- With the right features + reduction, **Transformer** based model (akin to CrabNet) will overcome performance of leaderboard Non-GNN models (RF, XGBoost, ML Ensembles)
- Stretch: **transfer learning** (pretrain on larger formation-energy data, then fine-tune) may help even more.

## Objective

- Predict **perovskite** formation energy (eV/cell) from available descriptors.

## Success Criteria

- Implement “simple baselines” (RF-SCM/Magpie **~0.23 MAE**).
- Aim to approach / beat best non-GNN leaderboard region (MODNet **~0.09 MAE**) with feasible compute.

# Leaderboard on perovskites

## Graph Neural Networks

| algorithm | mean mae | std mae | mean rmse | max max_error |
|---|---|---|---|---|
| coGN | 0.0269 | 0.0008 | 0.0554 | 0.9449 |
| ALIGNN | 0.0288 | 0.0009 | 0.0559 | 0.9028 |
| DeeperGATGNN | 0.0288 | 0.0011 | 0.0568 | 1.0267 |
| coNGN | 0.0290 | 0.0011 | 0.0590 | 0.9346 |
| Finder_v1.2 structure-based version | 0.0320 | 0.0012 | 0.0594 | 0.8875 |

## Featurizers + ML Regressor

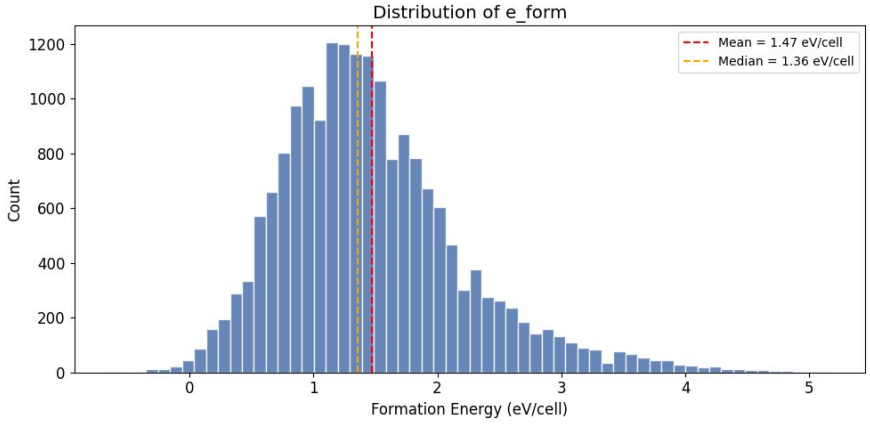
| algorithm | mean mae | std mae | mean rmse | max max_error |
|---|---|---|---|---|
| MODNet (v0.1.12) | 0.0908 | 0.0028 | 0.1277 | 1.1780 |
| MODNet (v0.1.10) | 0.0908 | 0.0028 | 0.1277 | 1.1780 |
| AMMExpress v2020 | 0.2005 | 0.0085 | 0.2954 | 3.3116 |
| RF-SCM/Magpie | 0.2355 | 0.0034 | 0.3346 | 2.8870 |

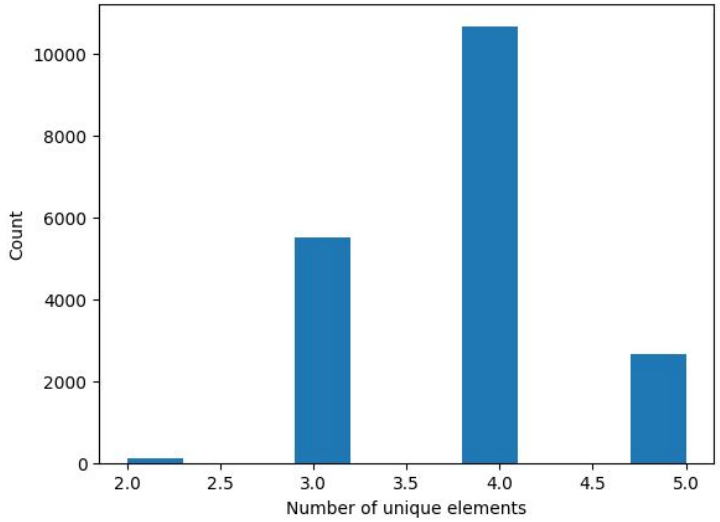
## Transformer (Composition Only)

| algorithm | mean mae | std mae | mean rmse | max max_error |
|---|---|---|---|---|
| CrabNet | 0.4065 | 0.0069 | 0.5412 | 2.3766 |

# Data Understanding: Datasets & Target

**Matbench perovskites** (~18,928 samples; target = formation energy, eV/cell)



# Data Understanding: Contents & Details

```
Full Formula (Re1 As1 O2 F1)
Reduced Formula: ReAsO2F
abc   :   4.292639   4.292639 4.292639
angles:  90.000000  90.000000  90.000000
pbc   :       True       True       True
Sites (5)
  #  SP           a    b    c
---  ----  --------  ---  ---
  0  Re    0.141617  0    0
  1  As    0.509386  0.5  0.5
  2  F     0.531687  0    0.5
  3  O     0.307487  0.5  0
  4  O     0.927582  0.5  0.5

e_form (target) = 1.48 eV/cell
```

```
df_raw.loc[sample_idx]
```

[11] ✓ 0.0s

...

| | 2 |
|---|---|
| structure | [[0.60790913 0. 0. ] Re, [2.186... |
| e_form | 1.48 |

- 'structure' is a pymatgen **Structure** obj:
  - lattice constants (abc)
  - angles
  - periodic boundary conditions
  - fractional coordinates

    … of each atomic site.

- 'e_form' is the DFT formation energy we want to predict.

# Data Prep: Feature Engineering Strategy

**Composition-only features**

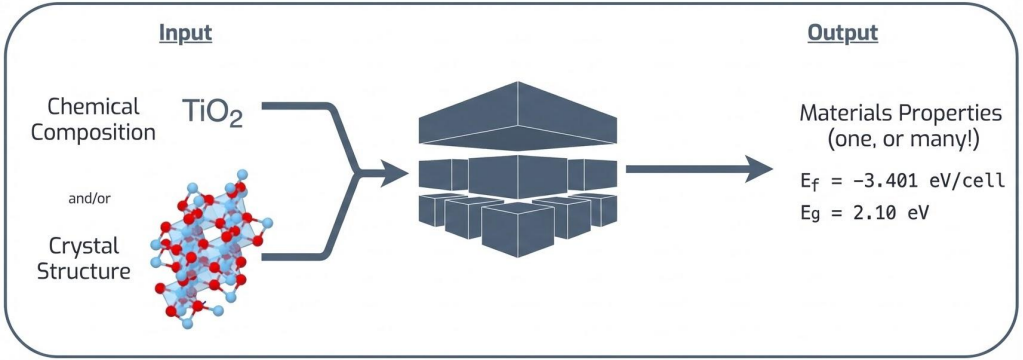
Magpie / elemental stats

Original CrabNet used only those

**Structure-aware features**

e.g., **Sine Coulomb Matrix (SCM)**, structural descriptors

MODNet / matminer structure featurizers (candidate pool)



# Composition Features: Magpie / elemental statistics

*Can generate up to 132 features from composition alone.*

**Stoichiometry / diversity**

- n_elements (how many different elements)
- fractional_composition_entropy
- max_atomic_fraction, min_atomic_fraction

ReAsO2F

**Statistics of elemental properties**

For each elemental property:

- p_mean, p_std, p_min, p_max, p_range

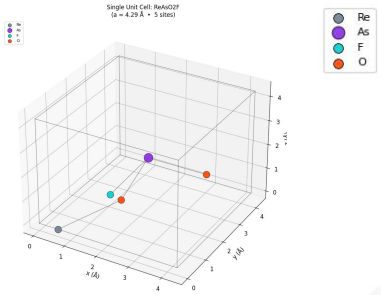
**Example properties**

- atomic_number_mean, atomic_number_std
- atomic_weight_mean
- electronegativity_mean, electronegativity_range
- covalent_radius_mean
- electron_affinity_mean
- melting_point_mean
- valence_electrons_mean
- d_block_fraction, p_block_fraction, s_block_fraction

# Structure Features: geometry + lattice + local environment

## A) Lattice / cell geometry

- volume, density
- a, b, c, alpha, beta, gamma
- packing_fraction
- num_sites / n_atoms

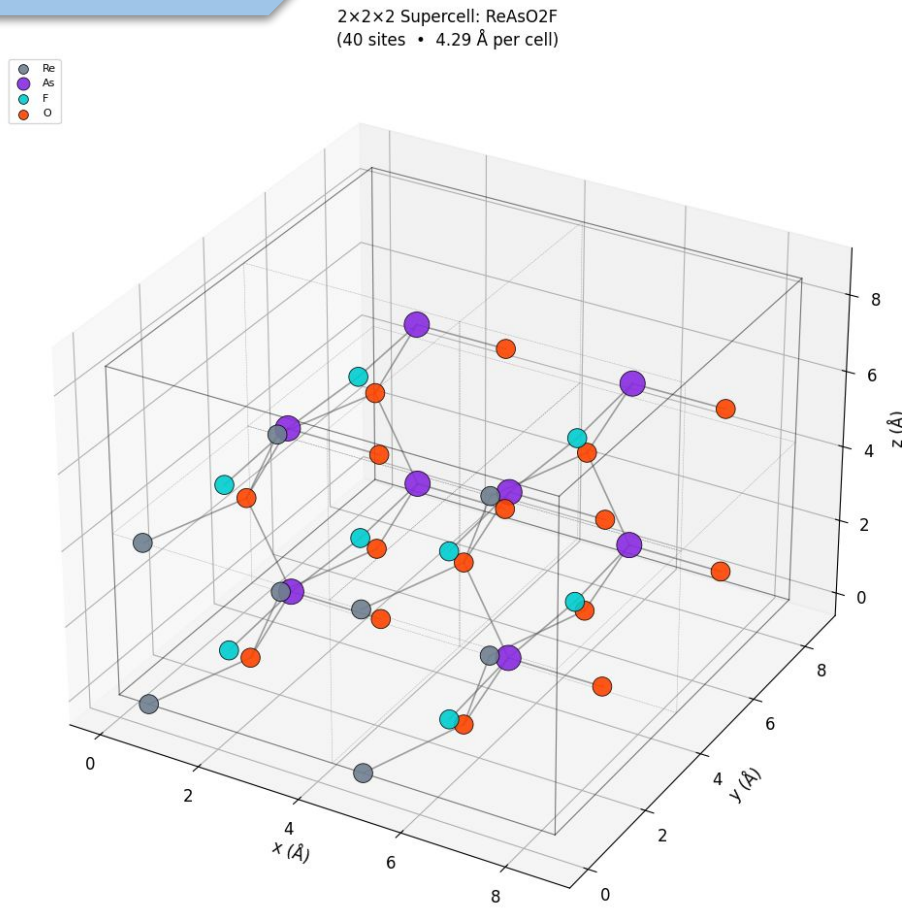



## B) Radial / distance statistics

- mean_neighbor_dis, std_neighbor_dis
- min_interatomic_dis, max_interatomic_dis
- coordination_number_mean

## C) Bonding / local environment

- avg_bond_length, bond_length_std
- avg_coord_A_site, avg_coord_B_site
- voronoi_coordination_stats



# Data Prep: Cleaning + Reduction

- NaN handling
- Imputation
- Encoding

200+ Features possible after featurization steps

- Correlation
- PCA
- Model-based methods
- Relief-based methods

Reduce to:
- Keeps training feasible

- Reduces overfitting risk on 18k samples with huge feature vectors

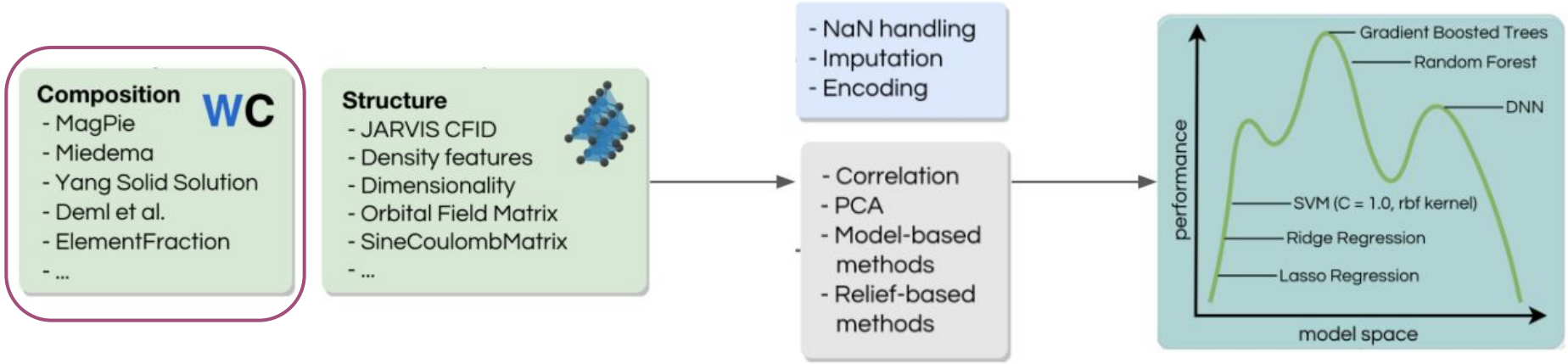
# Modeling: Baselines

**Baseline A (structure+composition):**

- **RF** on SCM (structure) + Magpie (composition)
- Result: **~0.23 MAE, 0.16 MAPE**
- **XGBoost** on same. Res: **~0.22 MAE, 0.14 MAPE**

**Baseline B (composition-only):**

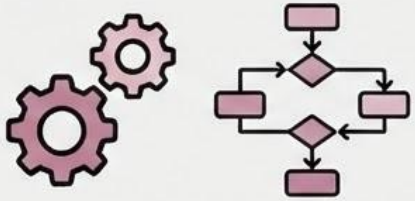
- **RF** gave **~0.53 MAE, 0.36 MAPE** (bad; negative $R^2$)
- GridSearch found **XGBoost** improved to **~0.43 MAE (0.29 MAPE)**



# Modeling: Target approach

**Goal:** “Beat non-GNN solutions using structure-aware features + smarter models.”

## MODNet-style approach



Ensemble + optimized feature pipeline as inspiration, also heavy hyperparameter tuning.

## Graph Neural Networks

Exploring SOTA approaches feasible for us to try.

## “Transformer on features” experiment & Stretch

Treat engineered feature vectors as tokens/groups. Compare vs pure MLP baseline.

### Stretch

Pretrain on mp_e_form, then fine-tune on perovskites (transformer or embeddings).

# Evaluation: Protocol + Metrics

## Protocol & Reporting

### Protocol (Matbench 5-fold CV)

Follow Matbench 5-fold CV

### Reporting

Mean ± std across folds. Also show max_error (optional, helps interpret "worst-case failures").

## Key Metrics

**MAE** (primary)

**RMSE**

**$R^2$**

**MAPE**

## Model Performance

| model | MAE | RMSE | $R^2$ |
|---|---|---|---|
| CHGNet | 0.07 | 0.11 | 0.61 |
| M3GNet | 0.07 | 0.12 | 0.59 |
| MEGNet | 0.13 | 0.20 | -0.27 |
| CGCNN | 0.14 | 0.23 | -0.61 |
| CGCNN+P | 0.11 | 0.18 | 0.02 |
| Wrenformer | 0.10 | 0.18 | -0.04 |
| BOWSR + MEGNet | 0.11 | 0.16 | 0.15 |
| Voronoi RF | 0.14 | 0.21 | -0.32 |
| dummy | 0.12 | 0.18 | 0.00 |

# What We Learned + Next Steps

**Current Status**

- Implemented Matbench protocol + baseline pipeline.
- Verified structure features matter (composition-only struggles).

**Next Steps (Concrete)**

- Expand structure featurizers (MODNet/matminer/autominer candidates).
- Compare XGBoost vs RF vs ensembles on hybrid feature sets
- Add consistent preprocessing + reduction + tuning.
- Stretch: pretraining transfer experiment if time allows.

**Risks**

- Feature leakage / fold contamination (mitigated via strict Matbench CV).
- Overfitting on high-dimensional descriptors.
- Feature computation cost vs model speed tradeoff.

# Contribution

**Dataset Understanding:**

Arukhan Kenes, Alisher Kalabayev

**Dataset Preparation/Feature Extraction:**

Aruzhan Suleimenova, Yernar Sakenov

**Modeling:**

Batyr Bodauday, Nurzhigit Askaraliyev

*Team Leading: Batyr Bodaubay*
*Moral & Financial support: Arukhan Kenes*
*Vibe manager: Yernar Sakenov*
*Clash Royalist: Nurzhigit Askaraliyev*
*Compute provider: Alisher Kalabayev*
*Coffee enjoyer: Aruzhan Suleimenova*

# References

MatBench

- https://matbench.materialsproject.org/
- https://github.com/materialsproject/matbench/tree/main/benchmarks/matbench_v0.1_rf
- https://github.com/materialsproject/matbench/tree/main/benchmarks/matbench_v0.1_CrabNet
- https://ml.materialsproject.org/projects/matbench_perovskites